

Existing Systems and Prior Approaches

Primary project source: ProofRoad v2, SIH 2026 PS 26007, uploaded by the user. The document describes ProofRoad as an engineering proposal and explicitly says it is not a safety certification. Page references below refer to the uploaded ProofRoad v2 document.

1. System categories relevant to PS 26007

System category	Typical role	Limitation in dense fog / mine use
Human visual operation	Direct driving judgement	Visibility collapses; conservative slowdown or stopping
RGB computer vision	Object/road recognition	Performance degrades strongly with dense fog
Thermal imaging	Warm-object detection	Useful for people/engines but range and resolution remain limited
NIR imaging	Night/dust contrast	Scattering in dense fog remains a limitation
LiDAR	3D ranging	ProofRoad excludes it for dense fog due to degradation/cost considerations
mmWave radar	Range and relative motion	Poor object naming and angular resolution; strong candidate for primary ranging
V2V/V2I	Share hazards/traffic state	Radio can fail or be jammed; cannot replace local sensing
Fog/visibility posts	Roadside visibility measurement	Sensor can drift or become dusty; needs audit
Central dashboard	Monitoring/replay	Should not be in the truck safety loop

2. What ProofRoad combines

The proposal combines radar-first perception, one-way sensor fusion, stopping-distance-based safe-speed advice, fail-closed gates, cooperative hazard sharing, freshness-aware messages, a dedicated safety MCU, and a windshield-first driver interface.

Its central architectural rule is that external information can add caution but cannot make the truck less conservative than its own radar/safety loop.

3. Prior designs synthesized in ProofRoad

ProofRoad states that it synthesizes six original solution designs plus AegisMesh and FogNet Sentinel. It identifies contributions such as stale-message rejection, safety black box, fog-post auditing, shared hazard records, proven-clear corridor calculations, measured braking bins, signed age-checked messages, dedicated safety MCU, Phantom Obstacle presentation and driver screen layout (p. 15).

The proposal deliberately drops timer-based hazard clearing, unauthenticated Stop All messages, percentage trust scores, large central-broker scope, and several future-work features.